Mathematical Modeling Lecture Notes
Queuing

I. Introduction
   a. We wait in lines all the time.
      i. The text says that Americans spend an average of 37 billion hours per year waiting in queues.
      ii. This comes to about 123 hours, or over five days, per person per year just waiting in line.
   b. How do you design an efficient queue?
      i. If you don't have enough servers then the lines get too long, people get mad, and leave.
      ii. If you have too many servers you end up paying the servers just to sit around.
      iii. Variability in service times is an issue, too.
   c. Basic model, graphically:

II. Background theory
   a. The Poisson process
      i. Definition
         1. A *counting process* $N(t)$ counts the number of events of a particular type that have occurred.
         2. A *Poisson process* with parameter $\lambda > 0$ is a counting process satisfying the following conditions:
            - $N(0) = 0$.
            - The number of events occurring in nonoverlapping intervals are independent.
            - The number of events occurring in a given interval depend on that interval only through its length.
            - $P(N(h) = 1) = \lambda h + o(h)$. In other words, the probability of exactly one event in a sufficiently short interval of length h is approximately λh.
            - $P(N(h) \geq 2) = o(h)$. In other words, the probability of two or more events in a sufficiently short interval of length h is approximately zero.
         3. Recall that a function $f(x)$ is $o(x)$ if $\lim_{x\to\infty} \frac{f(x)}{x} = 0$.
      ii. Since events occurring in nonoverlapping intervals are independent, we often say that arrivals governed by a Poisson process occur randomly.
      iii. Find the probability of exactly x arrivals in an interval of length t.
         1. Divide the interval of length t into n subintervals of length t/n.
         2. If n approaches infinity, then the subinterval lengths get smaller and smaller, and the probability of having two or more events in any subinterval approaches 0.

3. So we can view the n subintervals as a sequence of n trials, where either one event occurs or no events occur.
   - We can call "one event occurs" a success, and "no events occur" a failure.
   - The probability of success is approximately $\lambda t/n$.
   - Thus we have, approximately, a binomial distribution with n trials and probability of success $\lambda t/n$.
4. So the probability of getting exactly x successes is approximately
$$\frac{n!}{(n-x)!x!}\left(\frac{\lambda t}{n}\right)^x\left(1-\frac{\lambda t}{n}\right)^{n-x}.$$
5. In the limit, we have
$$\lim_{n\to\infty}\frac{n(n-1)\cdots(n-x+1)}{n^x}\left(\frac{(\lambda t)^x}{x!}\right)\left(1-\frac{\lambda t}{n}\right)^n\left(1-\frac{\lambda t}{n}\right)^{-x}$$
$$=(1)\left(\frac{(\lambda t)^x}{x!}\right)e^{-\lambda t}(1)$$
$$=\frac{(\lambda t)^x}{x!}e^{-\lambda t}.$$
6. Thus the probability of having exactly x arrivals in an interval of length t is
$$P(N(t)=x)=\frac{(\lambda t)^x e^{-\lambda t}}{x!}.$$

iv. Interarrival times
1. Let T_i denote the ith interarrival time; that is, the time between arrivals $i-1$ and i.
2. If we assume a Poisson process, what's the distribution of T_i?
   - If S_i is the time of the ith arrival, then $P(T_i > t) = P(N(S_{i-1}+t) - N(S_{i-1}) = 0)$.
   - However, since the number of events occurring in an interval depends only on the length of the interval, this probability is the same regardless of the value of i or of S_{i-1}.
   - Thus $P(T_i > t) = P(N(0+t) - N(0) = 0) = P(N(t) = 0) = e^{-\lambda t}$.
   - Therefore, $P(T_i \le t) = 1 - e^{-\lambda t}$.
   - And the probability density function of T_i is $f_{T_i}(t) = \lambda e^{-\lambda t}$, $t \ge 0$.
   - Thus $T_i \sim exponential(\lambda)$.

b. The exponential distribution

i. The expected value of an $exponential(\lambda)$ distribution is easy to derive (via integration by parts).
1. It's $1/\lambda$.
2. This makes sense: If the arrivals occur at a rate of λ, the interarrival times should, on average, be $1/\lambda$.

ii. The variance of an $exponential(\lambda)$ distribution is also easy to derive: It's $1/\lambda^2$.

iii. Memoryless property of the exponential distribution
1. This means that, given the current time, the distribution of the time to the next arrival does not depend on anything in the past.
2. In particular, it does not depend on how many arrivals have occurred, and it does not depend on how long it has been since the last arrival.
3. If arrivals really do occur randomly, this should make sense.

4. For service times, it make sense in some contexts, but not in others.
   - For example, for some services the customers can want very different things, producing a wide variability in service times.
   - The fact that the customer has been with the server for quite some time already may simply mean that this customer needs quite a bit of help and may not give any additional information into how much longer the customer will be with the server.
   - However, for some services the variability in service times can be quite small or even approximately zero.
   - In these situations the exponential distribution doesn't fit as well.

III. Terminology and basic concepts
   a. Queues are described as `_/_/_`.
      - The blanks represent, respectively, the distribution of interarrival times, the distribution of service times, and the number of servers.
   b. Arrivals
      i. Arrivals are generally modeled by a Poisson process with rate λ.
      ii. We don't know exactly how many customers will arrive in a given time period (hour), but the rate is λ (per hour).
      iii. The Poisson model implies that the interarrival times are memoryless (as we have just shown).
      iv. If the arrivals follow a Poisson process the queue is an `M/_/_` queue.
         - The M is for Markovian, because of the memoryless interarrival times.
   c. Service
      i. Number of servers
         1. Single-server models
         2. Multiple-server models
      ii. There are multiple commonly-used models for the distribution of service times for a single server.
         1. The exponential is one such model. The notation here is M, again for Markovian.
         2. Some servers, especially automated servers, are assumed to have constant times. The notation here is D, for degenerate.
         3. Sometimes the server has to perform the same sequence of tasks, each of which has the exponential distribution.
            - So the service time ends up being the sum of several exponentially distributed random variables.
            - If these are independent and identically distributed then the total service time has a gamma distribution.
            - However, in queuing this is called an *Erlang distribution* with parameter k, where k is the number of tasks.
            - The notation here is E_k.
         4. You can also have a G, which denotes a general distribution.
      iii. Whatever the distribution, the service rate is denoted μ.
   d. Measures of performance
      i. The basic measures of performance of a queueing system are the following:
         1. L_q, the expected number of customers in the queue

2. L, the expected number of customers in the system (in the queue plus those being served)
3. W_q, the expected time a customer spends waiting in the queue
4. W, the expected time a customer spends waiting in the system
5. P_n, the probability that there are exactly n customers in the system

ii. Relationships

1. $W = W_q + 1/\mu$
2. Little's formula
   - Thought experiment: If $W = 5$ (waiting time of 5 hours on average for the queue), and $\lambda = 4$ (4 arrivals per hour), then the expected queue length should be $L = 20$ customers.
   - Generalizing, we have *Little's formula*: $L = \lambda W$.
   - This holds when the arrival rate is constant.
   - If the arrival rate is not constant, then Little's formula holds for the *mean* arrival rate.
   - More generally, λ is the long-term average effective arrival rate. (In this situation, sometimes we use $\bar{\lambda}$ instead of λ to indicate the difference.)
3. Similarly, $L_q = \lambda W_q$.
4. Putting these two relationships together we have $L = \lambda(W_q + 1/\mu) = \lambda W_q + \lambda/\mu = L_q + \lambda/\mu$.
5. By definition, $L = \sum_{n=0}^{\infty} nP_n$.
6. Similarly, $L_q = \sum_{n=s}^{\infty}(n-s)P_n$.

IV. More on the Poisson process and the exponential distribution

a. Aggregation of Poisson processes

i. Suppose that there are n different types of customers.

ii. Suppose type i arrives according to a Poisson process with rate λ_i.

iii. Suppose also that the arrivals of customers of type i are independent of the arrivals of customers of type j.

iv. If $N_i(t)$ is the number of customers of type i who arrive to the system in a time period of length t, then let $N(t) = \sum_{i=1}^{n} N_i(t)$ be the total number of customers who arrive to the system in a time period of length t.

v. We know that $N_i(t) \sim Poisson(\lambda_i t)$.

vi. What is the distribution of $N(t)$?

1. $N_i(t)$ has moment-generating function $M(s) = \exp(\lambda_i t(e^s - 1))$.
2. By the properties of mgf's,

$$\begin{aligned} M_{N(t)}(s) &= \prod_{i=1}^{n} \exp(\lambda_i t(e^s - 1)) \\ &= \exp\left(\sum_{i=1}^{n} \lambda_i t(e^s - 1)\right) \\ &= \exp\left((e^s - 1)\left(\sum_{i=1}^{n} \lambda_i t\right)\right), \end{aligned}$$

which is the moment-generating function of a Poisson random variable with mean $t\sum_{i=1}^{n} \lambda_i$.

3. Therefore, the total number of customers in a time period of length t is Poisson$(t\sum_{i=1}^{n}\lambda_i)$.

vii. Aggregation of Poisson processes is still Poisson.

b. Disaggregation of Poisson processes

i. Suppose each customer has a fixed probability, p_i, of being of type i.

- What is the probability that there are exactly $N_i(t)$ customers of type i arriving in a time period of length t?

$$\begin{aligned}
P(N_i(t)=y) &= \sum_{x=y}^{\infty} P(N_i(t)=y|N(t)=x)P(N(t)=x) \\
&= \sum_{x=y}^{\infty}\binom{x}{y}(p_i)^y(1-p_i)^{x-y}\frac{(\lambda t)^x e^{-\lambda t}}{x!} \\
&= e^{-\lambda t}(p_i)^y\sum_{x=y}^{\infty}\frac{x!}{y!(x-y)!}(1-p_i)^{x-y}\frac{(\lambda t)^x}{x!} \\
&= \frac{e^{-\lambda t}(p_i)^y}{y!}\sum_{z=0}^{\infty}\frac{(1-p_i)^z(\lambda t)^{z+y}}{z!} \\
&= \frac{e^{-\lambda t}(p_i)^y(\lambda t)^y}{y!}\sum_{z=0}^{\infty}\frac{((1-p_i)\lambda t)^z}{z!} \\
&= \frac{e^{-\lambda t}(p_i)^y(\lambda t)^y}{y!}e^{(1-p_i)\lambda t} \\
&= \frac{e^{-p_i\lambda t}(p_i\lambda t)^y}{y!}.
\end{aligned}$$

- Thus $N_i(t) \sim Poisson(p_i\lambda t)$.
- Disaggregation of Poisson processes is still Poisson.

ii. Another use of disaggregation with Poisson processes is to model *balking*, that is, customers leaving without entering the queue.

- If the arrivals to the system follow a Poisson process with parameter λ, and each customer has a fixed probability p of balking, then the arrival of customers who actually enter the queue follows a Poisson process with parameter $p\lambda$.

c. Clustered arrivals

i. What's the probability of having an interarrival time longer than $1/\lambda$?

ii. This is $P(T_i > 1/\lambda) = \int_{1/\lambda}^{\infty}\lambda e^{-\lambda t}dt = -e^{-\lambda t}|_{1/\lambda}^{\infty} = 0 + e^{-1} \approx 0.368$.

iii. So only 36.8% of the interarrival times will be longer than the mean.

iv. You're more likely to see short arrival times than long arrival times.

v. So truly random arrivals tend to cluster, rather than be spread out evenly.

vi. Graphically:

d. Minimums of exponential random variables

i. Let $T_i \sim exponential(\alpha_i)$, and let the T_i's be independent.

ii. Let $T = \min_i T_i$. What is the distribution of T?

iii. For example, suppose you have n servers, each of which is currently working with a customer. What is the distribution of the time until the first customer leaves?

iv. It turns out that $T \sim exponential\left(\sum_{i=1}^{n} \alpha_i\right)$.

$$\begin{aligned} P(T > t) &= P(T_1 > t, T_2 > t, \ldots, T_n > t) \\ &= P(T_1 > t)P(T_2 > t) \cdots P(T_n > t) \\ &= \prod_{i=1}^{n} e^{-\alpha_i t} \\ &= \exp\left(-t \sum_{i=1}^{n} \alpha_i\right). \end{aligned}$$

v. This also means that if the servers have identical mean service rates α, then the mean rate for the entire system is $n\alpha$.

vi. Moreover, because of memorylessness, the distribution of interservice times between customers with different servers is $exponential(n\alpha)$.

vii. So, in the Markovian model, a system with n servers but identical mean service rates α can sometimes be treated as a single-server system with mean service rate $n\alpha$.

e. Birth-and-death processes

i. If we assume interarrival times and service times for a queue are exponentially distributed, then the queue can be modeled as a special kind of continuous-time Markov chain called a *birth-and-death process*.

ii. In addition to the continuous-time Markov chain assumptions, the birth-and-death process assumes the following:

1. The state of the system is the number of individuals in the system.
2. If the system is in state n, then the transition rate to state $n + 1$ is λ_n, $n \geq 0$, and the transition rate to state $n - 1$ is μ_n, $n \geq 1$.
3. Thus λ_n is the birth rate for state n, and μ_n is the death rate for state n.

iii. Pictorially:

iv. Steady-state probabilities

1. We can determine these in the same manner that we determine them for any continuous-time Markov chain.
2. With respect to queuing notation, these are the P_n's.
3. As we must have "rate in = rate out" for each state, we have the following set of equations:
   - $\lambda_0 P_0 = \mu_1 P_1$
   - $(\lambda_1 + \mu_1)P_1 = \lambda_0 P_0 + \mu_2 P_2$
   - In general, $(\lambda_n + \mu_n)P_n = \lambda_{n-1} P_{n-1} + \mu_{n+1} P_{n+1}$

4. This produces

$$P_1 = \frac{\lambda_0}{\mu_1} P_0$$

$$P_2 = \frac{(\lambda_1 + \mu_1)P_1 - \lambda_0 P_0}{\mu_2} = \frac{(\lambda_1 + \mu_1)P_1 - \mu_1 P_1}{\mu_2} = \frac{\lambda_1}{\mu_2} P_1 = \frac{\lambda_1 \lambda_0}{\mu_2 \mu_1} P_0.$$

5. Now, suppose $P_n = \frac{\lambda_{n-1}}{\mu_n} P_{n-1}$. Then

$$P_{n+1} = \frac{(\lambda_n + \mu_n)P_n - \lambda_{n-1} P_{n-1}}{\mu_{n+1}} = \frac{(\lambda_n + \mu_n)P_n - \mu_n P_n}{\mu_{n+1}} = \frac{\lambda_n}{\mu_{n+1}} P_n.$$

6. This immediately implies

$$P_n = \frac{\prod_{i=0}^{n-1} \lambda_i}{\prod_{i=1}^{n} \mu_i} P_0.$$

7. Therefore,

$$\sum_{n=0}^{\infty} \frac{\prod_{i=0}^{n-1} \lambda_i}{\prod_{i=1}^{n} \mu_i} P_0 = 1 \implies P_0 = \left(\sum_{n=0}^{\infty} \frac{\prod_{i=0}^{n-1} \lambda_i}{\prod_{i=1}^{n} \mu_i} \right)^{-1}.$$

v. Other measures of performance
   1. With the P_n's we are in a position to determine these.
   2. We can find L and L_q via $L = \sum_{n=0}^{\infty} n P_n$ and $L_q = \sum_{n=s}^{\infty} (n-s) P_n$.
   3. Using $W = L/\lambda$ is a bit trickier, though, if the λ_i's aren't the same.
      - Let $\bar{\lambda}$ be the overall mean arrival rate.
      - Thus, in terms of the λ_i's, $\bar{\lambda} = \sum_{n=0}^{\infty} \lambda_n P_n$.
      - And then the actual formula to use is $W = L/\bar{\lambda}$.

f. Example (Problem 6.13 in Ross, 5th ed.): A small barbershop, operated by a single barber, has room for at most two customers. Potential customers arrive at a Poisson rate of three per hour, and the successive service times are independent exponential random variables with mean 1/4 hour.
   i. We have $\lambda_0 = \lambda_1 = 3$, $\lambda_i = 0$ for $i \geq 2$.
   ii. We also have $\mu_1 = \mu_2 = 4$, $\mu_i = 0$ for $i \geq 3$.
   iii. The state transition diagram:

   iv. What is the average number of customers in the shop?
      1. This is $0P_0 + 1P_1 + 2P_2$.
      2. We need to solve the following system:

$$3P_0 = 4P_1,$$
$$3P_1 = 4P_2.$$

      3. We get $P_1 = \frac{3}{4} P_0$, and $P_2 = \frac{3}{4} P_1 = \frac{9}{16} P_0$.
      4. Normalizing, we have $P_0 = 16/37, P_1 = 12/37, P_2 = 9/37$.
      5. Therefore, the average number of customers in the shop is $12/37 + 18/37 = 30/37$.
   v. What is the proportion of potential customers that enter the shop?

- This is $1 - P_2 = 28/37 \approx 0.76$.

vi. If the barber could work twice as fast, how much more business would he do?

1. If we double μ to 8 per hour, then we need to solve the system

$$3P_0 = 8P_1,$$
$$3P_1 = 8P_2.$$

2. We get $P_1 = \frac{3}{8}P_0$, and $P_2 = \frac{3}{8}P_1 = \frac{9}{64}P_0$.
3. Normalizing, we have $P_0 = 64/97, P_1 = 24/97, P_2 = 9/97$.
4. Now he serves $1 - P_2 = 0.91$ of the potential customers.
5. This is an increase of 15% of the total.
6. Put another way, he loses less than half as many customers as he did before.

V. Queues fitting the birth-and-death process model

a. The `M/M/s` model

i. The rate diagram looks like the following:

ii. Derivations of measures of performance

1. We know $\lambda_n = \lambda$ for all n.
2. $\mu_n = \begin{cases} n\mu, & n \le s; \\ s\mu, & n > s. \end{cases}$
3. Thus

$$\begin{aligned} P_0 &= \left(\sum_{n=0}^{s-1} \frac{\prod_{i=0}^{n-1} \lambda}{\prod_{i=1}^{n} i\mu} + \sum_{n=s}^{\infty} \frac{\prod_{i=0}^{n-1} \lambda}{\left(\prod_{i=1}^{s} i\mu\right)\left(\prod_{i=s+1}^{n} s\mu\right)} \right)^{-1} \\ &= \left(\sum_{n=0}^{s-1} \frac{\lambda^n}{\mu^n n!} + \sum_{n=s}^{\infty} \frac{\lambda^n}{s!\mu^s (s\mu)^{n-s}} \right)^{-1} \\ &= \left(\sum_{n=0}^{s-1} \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!} + \left(\frac{\lambda}{\mu}\right)^s \frac{1}{s!} \sum_{n=s}^{\infty} \left(\frac{\lambda}{s\mu}\right)^{n-s} \right)^{-1} \\ &= \left(\sum_{n=0}^{s-1} \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!} + \left(\frac{\lambda}{\mu}\right)^s \frac{1}{s!} \frac{1}{1 - \lambda/(s\mu)} \right)^{-1}. \end{aligned}$$

4. When $s = 1$ this reduces to

$$\begin{aligned} P_0 &= \left(1 + \left(\frac{\lambda}{\mu}\right) \frac{1}{1 - \lambda/\mu}\right)^{-1} = \left(1 + \frac{\lambda}{\mu - \lambda}\right)^{-1} = \left(\frac{\mu - \lambda}{\mu - \lambda} + \frac{\lambda}{\mu - \lambda}\right)^{-1} \\ &= \left(\frac{\mu}{\mu - \lambda}\right)^{-1} = \frac{\mu - \lambda}{\mu} = 1 - \frac{\lambda}{\mu}. \end{aligned}$$

5. In the s server case we have $P_n = \begin{cases} \dfrac{\lambda^n}{n!\mu^n} P_0, & n \le s; \\ \dfrac{\lambda^n}{s! s^{n-s} \mu^n} P_0, & n > s. \end{cases}$

6. Thus we can derive an expression for L_q:

$$L_q = \sum_{n=s}^{\infty}(n-s)P_n = \sum_{i=0}^{\infty} iP_{s+i} = \sum_{i=0}^{\infty} i\frac{\lambda^{i+s}}{s!s^i\mu^{i+s}}P_0 = \frac{\lambda^{s+1}}{s!(s)\mu^{s+1}}P_0\sum_{i=0}^{\infty} i\left(\frac{\lambda}{s\mu}\right)^{i-1}$$
$$= \frac{\lambda^s}{s!\mu^s}\rho P_0\sum_{i=0}^{\infty}\frac{d}{d\rho}\rho^i, \text{ where } \rho = \lambda/(s\mu) \text{ and is called the } \textit{utilization factor}$$
$$= \frac{\lambda^s}{s!\mu^s}\rho P_0\frac{d}{d\rho}\sum_{i=0}^{\infty}\rho^i = \frac{\lambda^s}{s!\mu^s}\rho P_0\frac{d}{d\rho}\frac{1}{1-\rho} = \left(\frac{\lambda}{\mu}\right)^s\frac{\rho P_0}{s!(1-\rho)^2}.$$

7. The previous calculation assumes that $\lambda/(s\mu) < 1$.
   - That should make sense, however.
   - If $\lambda/(s\mu) \geq 1$, then the customers are arriving faster than they can be served (on average), and L_q will grow without bound.
8. The other measures of performance can be calculated directly from L_q.

iii. Example: (17.6-11) The Security & Trust Bank employs 4 tellers to serve its customers. Customers arrive according to a Poisson process at a mean rate of 2 per minute. However, business is growing, and management projects that the mean arrival rate will be 3 per minute a year from now. The transaction time between the teller and customer has an exponential distribution with a mean of 1 minute. Management has established the following guidelines for satisfactory level of service to customers. The average number of customers waiting in line to begin service should not exceed 1. At least 95% of the time, the number of customers waiting in line should not exceed 5.

1. Use the $M/M/s$ model to determine how well these guidelines are currently being satisfied.
   - We have $\lambda = 2$ customers per minute, $\mu = 1$ customer per minute, and $s = 4$.
   - Note that $\mu = 1/1$ here (where the 1 in the denominator is the number given).
   - Using the formulas, we have $L_q = 0.17$ and $P_0 + P_1 + \cdots + P_9 = 0.997$.
   - Both guidelines are being satisfied.
2. Evaluate how well the guidelines will be satisfied a year from now if no change is made in the number of tellers.
   - Now $\lambda = 3$ customers per minute, and the other measures remain the same.
   - Via the formulas $L_q = 1.53$, and $P_0 + P_1 + \cdots + P_9 = 0.909$.
   - Neither of the guidelines is satisfied.
3. Determine how many tellers will be needed a year from now to completely satisfy these guidelines.
   - If $s = 5$ we have $L_q = 0.35$, and $P_0 + P_1 + \cdots + P_{10} = 0.989$.
   - Thus 5 tellers suffice.

b. Finite-length queues

i. The $M/M/s$ model assumes that the queue can take on infinite length.

1. This isn't realistic for some situations.
2. For instance, there may only be so much space in a fast-food restaurant drive-through.

ii. We use the terminology $M/M/s/K$, where K represents the maximum number of customers in the system.

iii. The arrival rates change; now we have $\lambda_n = \begin{cases} \lambda, & n < K; \\ 0, & n \geq K. \end{cases}$

iv. See handout for derivations of expressions for the measures of performance.

- An important thing to note is that the infinite sum in the expression for P_0 is now finite.

c. Queues with finite calling populations

i. Another variation of the $M/M/s$ model allows for the population to be finite.

1. The machine repair problem that we considered when we discussed continuous-time Markov chains actually fits this situation.
2. The state of the system would be the number of machines broken down.

ii. If the maximum population is N then the state transition diagram looks like this:

iii. This assumes that the distribution of time an individual spends outside the system is also exponential.

iv. See handout for derivations of expressions for the measures of performance.

- Once again, the infinite sum for P_0 is now finite.

VI. Queues with nonexponential distributions

a. We're not going to talk about the cases in which the arrivals are not governed by a Poisson process, but you can read about some of these in the text.

b. General service times: The $M/G/s$ model

i. If we don't know the distribution of service times it can be difficult to determine the measures of performance.

ii. However, in the $s = 1$ case we do have a few results:

1. $P_0 = 1 - \rho$, where ρ is the utilization factor, λ/μ.
2. $$L_q = \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)}$$
3. $L = \rho + L_q$ (talk about why this makes sense)

c. Constant service times: The $M/D/s$ model

i. For some queuing systems, particularly those with some automation, the service time is the same for every customer.

ii. In the case $s = 1$ we have

$$L_q = \frac{\rho^2}{2(1-\rho)}.$$

iii. The other measures can be derived from this.

d. Erlang service times: The $M/E_k/s$ model

i. For some queuing systems the service involves the same set of k sequential tasks.

1. If each task has the same completion rate, then the total time to complete the tasks has an $Erlang(k, \mu)$ distribution.

2. Here, μ is the completion rate for the entire set of tasks, so each individual task has completion rate $k\mu$.
3. For those of you with some probability background, this means that the Erlang distribution is a special case of the gamma distribution.

ii. The mean of the Erlang distribution is $1/\mu$, and the variance is $1/(k\mu^2)$.

iii. In addition to its interpretation as a model for service times involving the same set of sequential tasks, it represents a middle-ground distribution between the higher-variability exponential service times and the zero variability constant service times.

- In fact, by finding the best values of k and μ the Erlang distribution is often fit to some given set of data when one is trying to model it using a queue.

e. See handout for other kinds of models.

VII. Examples

a. 17.7-4: Marsha operates an espresso stand. Customers arrive according to a Poisson process at a mean rate of 30 per hour. The time needed by Marsha to serve a customer has an exponential distribution with a mean of 75 seconds.

i. Find L, L_q, W, and W_q.

1. This is the $M/M/1$ model.
2. We have $\lambda = 30$ and $\mu = 60(60)/75 = 48$, which means $\rho = 5/8 = 0.625$.
3. Thus $L = \lambda/(\mu - \lambda) = 30/(48 - 30) = 30/18 = 5/3 = 1.67$.
4. We then have $L_q = L - \rho = 5/3 - 5/8 = 1.04$.
5. Finally, $W = L/\lambda = 0.056$ and $W_q = L_q/\lambda = 0.035$.

ii. Suppose Marsha is replaced by an espresso vending machine that requires exactly 75 seconds for each customer to operate. Find L, L_q, W, and W_q.

1. This is the $M/D/1$ model, with $\lambda = 30$, $\mu = 48$, and $\rho = 5/8 = 0.625$.
2. We have

$$L_q = \frac{\rho^2}{2(1-\rho)} = \frac{(5/8)^2}{2(3/8)} = 25/48 \approx 0.521.$$

3. Thus $L = 0.521 + 0.625 = 1.146$, $W = 1.146/30 \approx 0.038$, and $W_q = 0.521/30 \approx 0.017$.

b. 17.7-5: Antonio runs a shoe repair store by himself. Customers arrive to bring a pair of shoes to be repaired according to a Poisson process at a mean rate of 1 per hour. The time Antonio requires to repair each individual shoe has an exponential distribution with a mean of 15 minutes.

i. Consider the formulation of this queuing system where the individual shoes (not pairs of shoes) are considered to be the customers. For this formulation, construct the rate diagram.

1. Rate diagram ($\lambda = 1, \mu = 4$):

2. This doesn't really fit any of our models.

ii. Now consider the formulation of this queuing system where the pairs of shoes are considered to be the customers. Identify the specific queuing model that fits this formulation.

- This is the $M/E_2/1$ model, with $\lambda = 1$ and $\mu = 2$.

iii. Calculate the expected number of pairs of shoes in the shop.
- We have $L = \frac{\lambda}{\mu} + \frac{1+k}{2k}\frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{1}{2} + \frac{3}{4}\frac{1}{2(1)} = \frac{1}{2} + \frac{3}{8} = \frac{7}{8}$.

iv. Calculate the expected amount of time from when a customer drops off a pair of shoes until they are repaired and ready to be picked up.
- This is $W = L/\lambda = \frac{7}{8}$ hour (52.5 minutes).

c. 7.6-28 (modified): George is planning to open a drive-through fast-food stand with a single service window that will be open approximately 200 hours per month in a busy commercial area. Space for a drive-through lane is available for a rental of \$200 per month per car length. George needs to decide how many car lengths of space to provide for his customers. Excluding this rental cost for the drive-through lane, George believes that he will average a profit of \$4 per customer served. He also estimates that customers will arrive randomly at a mean rate of 20 per hour, although those who find the drive-through lane full will be forced to leave. George assumes that the time required to serve a customer will have an exponential distribution with a mean of 2 minutes.

i. This is a $M/M/1/K$ model, with $\lambda = 20$ and $\mu = 30$ per hour.

ii. Formulas (from handout)

$$P_n = \frac{1-\rho}{1-\rho^{K+1}}\rho^n.$$

$$L = \frac{\rho}{1-\rho} - \frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}}$$

$$L_q = L - (1 - P_0)$$

iii. Unfortunately, we don't have $L = \lambda W$ and $L_q = \lambda W_q$ anymore because $\lambda_n = 0$ when $n = K$.
1. However, define $\bar{\lambda} = \sum_{n=0}^{\infty} \lambda_n P_n = \sum_{n=0}^{K-1} \lambda P_n = \lambda(1 - P_K)$.
2. Then we have $L = \bar{\lambda} W$ and $L_q = \bar{\lambda} W_q$.

iv. Find L and the percentage of customers lost when the number of car lengths of space provided is 2, 3, 4, and 5.
1. For two spaces:
- The percentage of customers lost is $P_2 = \frac{1-2/3}{1-(2/3)^3}(2/3)^2 = 0.211$.
- The value of L is $(2/3)/(1-2/3) - 3(2/3)^3/(1-(2/3)^3) = 0.736$.
2. Three spaces: $L = 1.016$, percentage of customers lost is $P_3 = 0.123$
3. Four spaces: $L = 1.242$, percentage of customers lost is $P_4 = 0.076$
4. Five spaces: $L = 1.423$, percentage of customers lost is $P_5 = 0.048$

v. Calculate W from L for the cases considered in the previous question.
1. Two spaces:
- We have $\bar{\lambda} = 20(1 - 0.211) = 15.78$.
- Then $W = 0.736/15.78 = 0.0467$.
2. Three spaces: $W = 0.0579$
3. Four spaces: $W = 0.0672$
4. Five spaces: $W = 0.0747$

vi. Use the results from the first question to calculate the increase in expected profit per hour (excluding space rental costs) when the number of car lengths of space provided is increased from 2 to 3, from 3 to 4, and from 4 to 5.

- The profit per hour for K spaces is $4\lambda(1 - P_K)$.

Spaces	Percent of Customers Lost	Expected Profit/Hour	Change
2	0.211	\$63.12	
3	0.123	\$70.16	\$7.04
4	0.076	\$73.92	\$3.76
5	0.048	\$76.16	\$2.24

vii. Compare the increases in expected profit found in the previous question with the cost per hour of renting each car length of space. What conclusion do you draw about the number of car lengths of space that George should provide?
   1. The cost of adding a car length is 200/200, or \$1 per hour.
   2. George should provide five car lengths, as the marginal profit for 3, 4, and 5 are all larger than \$1 per hour.

VIII. Priority queues
   a. *Priority queues* are queues in which the customers are divided into a certain number of priority classes, and customers in higher priority classes are always served before customers in lower priority classes.
   b. Some examples of priority queues
      i. Some airline ticket counters, where the classes are 1) first/business class, 2) coach
      ii. The emergency room at a hospital, where people with more serious conditions are treated first
   c. Two types of priority queues
      i. Nonpreemptive: When a high priority arrival occurs the customer currently being served has his/her service completed before the new arrival is served.
      ii. Preemptive: When a high priority arrival occurs the customer currently being served is sent back to the queue, and the new arrival is served instead.
   d. Performance of priority queues
      i. We generally assume that customers in class i arrive according to a Poisson process with rate λ_i and that the service times for all classes have the same distribution.
      ii. This implies that the overall arrival process is Poisson with rate $\sum_i \lambda_i$.
      iii. In the nonpreemptive model, the only thing in the service that changes is the order in which customers are selected.
         1. The service itself doesn't change.
         2. Thus, in terms of calculating the overall measures of performance, the $M/M/s$ model works.
      iv. In the preemptive model, when a customer is preempted the distribution of his/her remaining service time is the same as if he/she were just beginning service.
         1. This is because of the memoryless property of the exponential distribution.
         2. As in the nonpreemptive model, the overall measures of performance can be calculated using the $M/M/s$ model.
      v. However, the measures of performance for the different priority classes are different.
         1. Classes with higher priority will have smaller values of W, W_q, L, and L_q.
         2. See handout for formulas for these expressions.

3. In particular, the highest priority class in a preemptive queue will behave exactly like an $M/M/s$ queue with a λ_1 arrival rate and a μ service rate.

IX. Queuing networks

a. There are many situations in which customers have to pass through a series of queuing systems.
   i. For example, at the DMV you often stand in line for one service. Then after that service is completed you stand in line for another service, and so forth.
   ii. Another common situation is a series of operations that a product must pass through in a manufacturing plant.

b. The major theoretical result on queuing networks is the following: If a service facility has an infinite queue, Poisson arrivals with rate λ, s servers, and exponential service times with service rate μ for each server such that $\lambda < s\mu$, then the output process of the facility in steady-state is Poisson with rate λ.
   i. In other words, in steady-state, customers leave just like they arrive.
   ii. Explanation
      1. The fact that the rates are the same should make sense for the steady-state, as that's part of what steady-state means.
      2. The Poisson distribution should make some sense, too: If service times are memoryless, then it makes some sense that they would produce random (Poisson) arrivals.
      3. The handout gives a reference for the proof.
   iii. The major usefulness of this result is that it allows us to analyze several queuing systems in series as if they were separate queuing systems all having the same Poisson arrival rate λ.
   iv. Unfortunately, in practice many queuing systems in series do not have infinite queues.

c. The text also discusses *Jackson networks*.
   i. These are networks of m queuing systems in which:
      1. Each system has an infinite queue.
      2. Customers leaving queuing system i enter queuing system j with probability p_{ij} or leave the network with probability $q_i = 1 - \sum_{j=1}^{m} p_{ij}$.
      3. Customers arrive at queuing system i from outside the system according to a Poisson process with rate α_i.
      4. The service time at system i is exponential with parameter μ_i.
   ii. In steady-state, each facility behaves as an independent $M/M/s$ queue with arrival rate $\lambda_j = \alpha_j + \sum_{i=1}^{m} \lambda_i p_{ij}$.

d. Example (Winston, p. 1105): The last two things that are done to a car before its manufacture is complete are installing the engine and putting on the tires. An average of 54 cars per hour randomly arrive requiring these two tasks. One worker is available to install the engine and can service an average of 60 cars per hour. After the engine is installed, the car goes to the tire station and waits for its tires to be attached. Three workers can serve at the tire station. Each works on one car at a time and can put tires on a car in an average of 3 minutes. All service times are exponential. Determine the mean queue length at each work station and the total expected time a car spends waiting for service.
   i. We have $\lambda = 54$, $\mu_1 = 60$, $\mu_2 = 20$.
   ii. For station 1 ($M/M/1$):

1. $L_q = \lambda^2/(\mu_1(\mu_1 - \lambda)) = 54^2/(60(6)) = 8.1$ cars.
2. $W_q = L_q/\lambda = 8.1/54 = 0.15$ hours.

iii. For station 2: $(M/M/3)$:

1. $L_q = 7.35$ cars.
2. $W_q = 7.35/54 = 0.136$ hours.

iv. Thus the total expected time a car spends waiting for service is 0.286 hours, or about 17 minutes.

v. Note that the mean wait time in the queue for the station with 3 workers at 20 cars per hour is smaller than that for the station with one worker at 60 cars per hour.

1. One reason for this is that when a (rare) particularly long service time occurs, the three workers don't all get backed up as badly as the one worker does.
2. Ask students for other reasons?

Homework: Handout problems #2-2, 4-2, 5-5, 5-9 (on part b, just find the P_n values), 6-20ab, 6-29abcd, 7-6, 8-1.